

substantial challenges in water balance. Nevertheless, they actively transport *specific* solutes that they maintain at levels different from those in the ocean. For example, homeostatic mechanisms in the Atlantic lobster (*Homarus americanus*) maintain a magnesium ion (Mg^{2+}) concentration in hemolymph (circulatory fluid) of less than 9 mM (millimolar, or 10^{-3} mol/L), far below the 50 mM concentration of Mg^{2+} in their environment.

Two osmoregulatory strategies evolved among marine vertebrates that address the challenges of a strongly dehydrating environment. One is found among marine “bony fishes,” a group that includes ray-finned and lobe-finned fishes. The other is found in marine sharks and most other chondrichthyans (cartilaginous animals; see Concept 34.3).

Cod, shown in **Figure 44.3a**, and other marine bony fishes constantly lose water by osmosis. They balance water loss by drinking a lot of seawater. The excess salts ingested with seawater are eliminated through the gills and kidneys.

Like bony fishes, sharks have an internal salt concentration much lower than that of seawater. However, shark tissues have a high concentration of several other solutes, including urea and another organic molecule, trimethylamine oxide (TMAO). Because the total solute concentration is somewhat higher than 1,000 mOsm/L, water slowly *enters* the shark’s body by osmosis and in food (sharks do not drink).

The small influx of water into the shark’s body is disposed of in urine produced by the kidneys. The urine also removes some of the salt that diffuses into the shark’s body; the rest is lost in feces or is secreted from a specialized gland.

Freshwater Animals

The osmoregulatory problems of freshwater animals are the opposite of those of marine animals. The body fluids

of freshwater animals must be hyperosmotic because animal cells cannot tolerate salt concentrations as low as that of lake or river water. Since freshwater animals have internal fluids with an osmolarity higher than that of their surroundings, they face the problem of gaining water by osmosis. For the perch and many other freshwater animals, water balance relies on excreting large amounts of very dilute urine and drinking almost no water (**Figure 44.3b**). In addition, salts lost by diffusion and in the urine are replenished by eating and by taking up salt across their gills.

Salmon and other euryhaline fishes that migrate between fresh water and seawater undergo dramatic changes in osmoregulatory status. When living in rivers and streams, salmon osmoregulate like other freshwater fishes, producing large amounts of dilute urine and taking up salt from the dilute environment through their gills. When they migrate to the ocean, salmon acclimatize (see Concept 40.2). They produce more of the steroid hormone cortisol, which increases the number and size of specialized salt-secreting cells. These and other physiological changes enable salmon in salt water to excrete excess salt from their gills and produce only small amounts of urine—just like bony fishes that spend their entire lives in salt water.

Animals That Live in Temporary Waters

Extreme dehydration, or *desiccation*, is fatal for most animals. However, a few aquatic invertebrates that live in temporary ponds and in films of water around soil particles can lose almost all their body water and survive. These animals enter a dormant state when their habitats dry up, an adaptation called **anhydrobiosis** (“life without water”). Among the most striking examples are the tardigrades, or water bears, tiny invertebrates less than

▼ **Figure 44.3** Osmoregulation in marine and freshwater bony fishes: a comparison.

